Organic Sudoku

How many questions will you need to solve?

Questions
(A – Z)

Target Syllabus: NCERT Chemistry Part II
You will (probably) need to answer most of these to solve the sudoku!

How does this work?

1. From the grid, choose a space containing a blue letter to solve.
2. Find its corresponding question below with *that* letter beside it.
3. Solve the question correctly and obtain the number (1 to 9).
4. Put that into your grid box!

You don’t need to solve all questions, but the puzzle becomes easier the more questions are solved!

Questions start at the next page.

A Choose the correct option and put its number into the grid box.

On treating *N*-Ethylphthalimide with _____, we will get Ethanamine.

(1) Methane
(2) Aqueous NaOH
(3) Methene
(4) Ethyne
(5) KOH
(6) Ethene
(7) N-Methylmethanamine
(8) $LiAlH_4$
(9) $NaBH_4$

B Choose the correct option and put its number into the grid box.

The order of basic strength of ethyl substituted amines is:

(P - primary, S - secondary, T - tertiary)

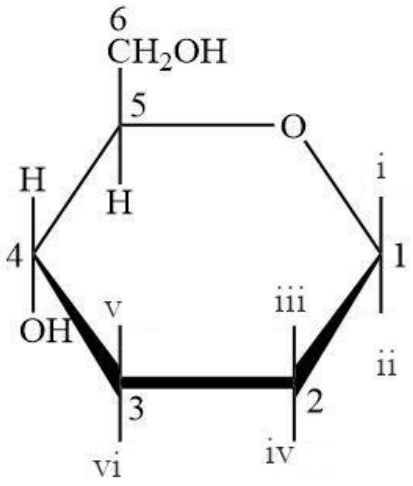
(1) P > S > T
(2) T > S > P
(3) S > T > P
(4) P > T > S
(5) T > P > S
(6) S > P > T

C Enter the number of letters of the answer into the grid box.

The common name of the organic product obtained from Fittig's Reaction is _____.

D Choose the correct option and put its number into the grid box.

The structure of β-D-(+)-glucopyranose is:



β-D-Glucose

(1) i-OH, ii-H, iii-H, iv-OH, v-H, vi-OH
(2) i-H, ii-OH, iii-H, iv-OH, v-H, vi-OH
(3) i-OH, ii-H, iii-OH, iv-H, v-H, vi-OH
(4) i-H, ii-OH, iii-OH, iv-H, v-H, vi-OH
(5) i-OH, ii-H, iii-H, iv-OH, v-OH, vi-H
(6) i-H, ii-OH, iii-H, iv-OH, v-OH, vi-H
(7) i-OH, ii-H, iii-OH, iv-H, v-OH, vi-H
(8) i-H, ii-OH, iii-OH, iv-H, v-OH, vi-H
(9) None of the above

E **Fill the blank with the correct option and put its number into the grid box.**

Using aldol condensation, we can react benzaldehyde with _____ to yield 1,3-Diphenylprop-2-en-1-one (Benzalacetophenone) as the major product.

(1) Itself
(2) Acetophenone
(3) Propiophenone
(4) Styrene

F **Choose the correct option and put its number into the grid box.**

A compound that reacts with Tollens' reagent but not Fehling's solution is:

(1) Acetone
(2) Acetaldehyde
(3) Formaldehyde
(4) Benzaldehyde

G **Choose the correct option and put its number into the grid box.**

When benzene is treated with CO, HCl in the presence of anhydrous $AlCl_3$ or CuCl, it gives:

(1) No reaction
(2) Benzaldehyde
(3) Benzoic Acid
(4) Salicylaldehyde
(5) Salicylic Acid
(6) Phenol
(7) Toluene
(8) Cumene
(9) Styrene

H **Choose the correct option and put its number into the grid box.**

The molar mass of the slow oxidation product of chloroform in the presence of light is:

a. divisible by 2
b. divisible by 3
c. divisible by 7
d. divisible by 11

Which statements are true?

(1) Only a and b
(2) Only a and c
(3) Only a and d
(4) Only b and c
(5) Only b and d
(6) Only c and d
(7) All but a
(8) All but b
(9) All but c

I **Choose the correct option and put its number into the grid box.**

Nitration of aniline yields around

(1) 51% *o*-nitroaniline, 47% *p*-nitroaniline, 2% *m*-nitroaniline
(2) 47% *o*-nitroaniline, 51% *p*-nitroaniline, 2% *m*-nitroaniline
(3) 57% *o*-nitroaniline, 2% *p*-nitroaniline, 41% *m*-nitroaniline
(4) 41% *o*-nitroaniline, 2% *p*-nitroaniline, 57% *m*-nitroaniline
(5) 2% *o*-nitroaniline, 41% *p*-nitroaniline, 57% *m*-nitroaniline
(6) 2% *o*-nitroaniline, 57% *p*-nitroaniline, 41% *m*-nitroaniline
(7) 2% *o*-nitroaniline, 47% *p*-nitroaniline, 51% *m*-nitroaniline
(8) 2% *o*-nitroaniline, 51% *p*-nitroaniline, 47% *m*-nitroaniline

J **Choose the correct option and put its number into the grid box.**

p-bromoacetanilide is hydrolysed to form a compound **X**, which then undergoes ammonolysis twice with benzyl chloride to form:

(1) N,N-dibenzyl-4-chloroaniline
(2) N-benzyl-4-chloroaniline
(3) N,N-dibenzylaniline
(4) N-phenylaniline
(5) N,N-diphenylaniline
(6) N-benzyl-4-bromoaniline
(7) N,N-dibenzyl-4-bromoaniline
(8) N-phenyl-4-bromoaniline
(9) N,N-diphenyl-4-bromoaniline

K **Enter the value of *x* from the given information.**

Acetophenone upon reacting with 2,4-dinitrophenylhydrazine forms its N-substituted derivative whose molecular formula is $\mathbf{C_aH_bN_cO_d}$.

Given: ***x* = 2*b - a - c - d**

(Hint: You do not need to find the derivative's structure)

L **Choose the correct option and put its number into the grid box.**

Other names for Thiamine and Pyridoxine respectively is

(1) Vitamin B_1, Vitamin B_2
(2) Vitamin B_1, Vitamin B_6
(3) Vitamin B_1, Vitamin C
(4) Vitamin B_2, Vitamin B_1
(5) Vitamin B_2, Vitamin B_6
(6) Vitamin B_2, Vitamin C
(7) Vitamin B_6, Vitamin B_1
(8) Vitamin B_6, Vitamin B_2
(9) Vitamin B_6, Vitamin C

M **Find the value of *x* and put it into the grid box.**
The deficiency of Vitamin B_x causes convulsions.

N **Choose the correct option and put its number into the grid box.**
Catalytic hydrogenation of carbon monoxide at high pressure and temperature (200-300 atm, 573-673 K) in the presence of $ZnO\text{-}Cr_2O_3$ catalyst yields an organic compound **Y**.
Vapours of **Y** are then passed over heated copper at 573 K and forms:

(1) Acetic acid
(2) Formic acid
(3) Acetaldehyde
(4) Formaldehyde
(5) Propanal
(6) Propanoic acid
(7) Benzaldehyde
(8) Benzoic acid

O **Choose the correct option and put its number into the grid box.**
Passing **$(CH_3CH_2)_3C–OH$** over heated copper at 573 K forms

(1) $(CH_3CH_2)_2C=CHCH_3$
(2) $(CH_3CH_2)_3C–CH=CH_2$
(3) CH_3COOH
(4) HCOOH
(5) CH_3CHO
(6) HCHO

P **Choose the correct option and put its number into the grid box.**
Catalyst(s) used to convert phenol to benzoquinone are:

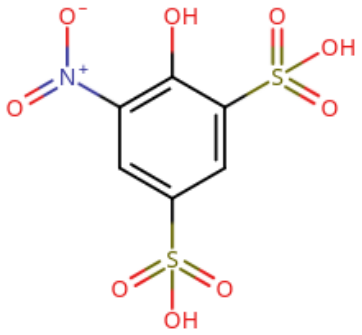
(1) PCC
(2) Jones Reagent
(3) $KMnO_4$
(4) Br_2
(5) HCl, CrO
(6) CO_2
(7) O_3
(8) Zn, H_2SO_4
(9) $Na_2Cr_2O_7$, H_2SO_4

Q **Find the value of *r* and put it into the grid box.**

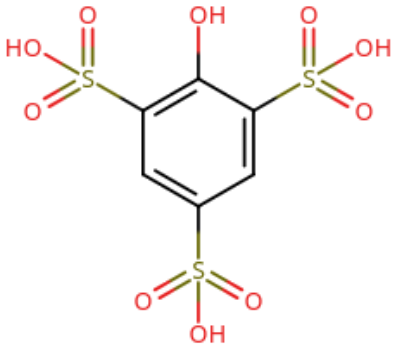
*"Nowadays, picric acid is prepared by treating phenol first to compound **Z**, and then with concentrated nitric acid."*

i. What is the structure of compound **Z**?
*(The option number chosen will be the value of **p**)*

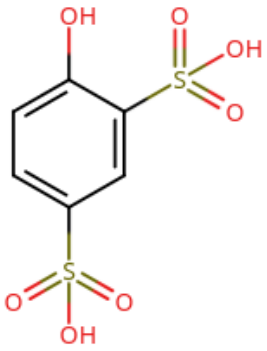
(1)



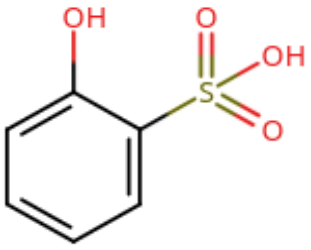
(2)



(3)



(4)



ii. If we directly use concentrated nitric acid on phenol to make picric acid, will the yield be poor?

*(The option number chosen will be the value of **q**)*

(1) No

(2) Yes

After you have found both of the above values, calculate the value of **r**:

r = p + 4q - 4.

R **Choose the correct option and put its number into the grid box.**
Hinsberg's reagent reacts with compounds L, M, N to form new organic compounds. Choose the correct option that describes the formed products from the reactions. Here, soluble and insoluble are relative to alkali.

L: Methanamide
M: N-methylmethanamide
N: N,N-dimethylmethanamide

(1) L: soluble, M: soluble, N: soluble
(2) L: insoluble, M: insoluble, N: insoluble
(3) L: soluble, M: insoluble, N: insoluble
(4) L: soluble, M: insoluble, N: insoluble
(5) L: insoluble, M: soluble, N: insoluble
(6) L: soluble, M: insoluble, N: soluble
(7) L: insoluble, M: soluble, N: soluble
(8) L: soluble, M: insoluble, N: does not form
(9) L: insoluble, M: soluble, N: does not form

S **Fill in the blank with the correct number and put it into the grid box.**
Compound **P**, an organic compound with _____ carbon atoms is oxidised in the presence of air and then treated with dilute acid to form carbolic acid and acetone.

T **Choose the correct option and put its number into the grid box.**
The reaction that takes place when (–)–2-Methylbutan-1-ol is heated with concentrated hydrochloric acid is an example of:

i. Inversion
ii. Retention
iii. Resonance

Which of the following statements are **NOT** true?
(1) None of the above
(2) i, ii, iii
(3) i
(4) ii
(5) iii
(6) i, ii
(7) ii, iii
(8) i, iii

U **Choose the correct option and put its number into the grid box.**

Which reaction results in benzaldehyde from toluene?

(1) Stephen reaction
(2) Etard reaction
(3) Rosenmund reaction

V **Choose the correct option and put its number into the grid box.**

Which reactant and product can be associated with Stephen reaction?

(1) Alcohol → Aldehyde
(2) Ketone → Aldehyde
(3) Ether → Aldehyde
(4) Ester → Aldehyde
(5) Alkene → Aldehyde
(6) Alkyne → Aldehyde
(7) Aldehyde → Aldehyde
(8) Acid chloride → Aldehyde
(9) Nitrile → Aldehyde

W **Choose the correct option and put its number into the grid box.**

Arrange the following in increasing boiling points:

a. Pentan-1-ol
b. n-Butane
c. Pentanal
d. Ethoxyethane

(1) d, b, c, a
(2) d, b, a, c
(3) b, d, c, a
(4) b, d, a, c
(5) d, c, b, a
(6) d, a, b, c
(7) b, c, d, a
(8) b, a, d, c

X **Choose the correct option and put its number into the grid box.**

Which of the following compounds has the highest boiling point?

(1) C_2H_5I
(2) C_2H_5Cl
(3) CH_3I
(4) CH_3Cl

Y **Choose the correct option and put its number into the grid box.**

Which of the following compounds has the highest boiling point?

(1) $CH_3CH_2CH_2CH_2Br$
(2) $CH_3CH_2BrCH_2CH_2$
(3) $(CH_3)_3CBr$

Z **Choose the correct option and put its number into the grid box.**

Which of the following compounds can undergo Cannizzaro reaction?

(1) Acetic acid
(2) Acetone
(3) Propanoic acid
(4) Acetophenone
(5) Acetaldehyde
(6) Benzaldehyde
(7) Two compounds from the above undergo the reaction.
(8) Three compounds from the above undergo the reaction.

Answer Key

(hidden in white text)

A
B
C

D
E
F
G
H

I
J
K

L
M
N
O
P
Q

R
S

T
U
V
W
X
Y
Z